

REPLICATION FROM WEBSITE:

Vehicle class:



ZERO40*

WORKS GoPro

22mm f/1.7 1/450s ISO500



			
2022	23,195 km	Petrol	Automatic

Registered In	Karachi	Color	York white
Assembly	Imported	Engine Capacity	1500 cc
Body Type	Crossover	Last Updated:	Mar 02, 2026

Ad Ref # 11196661

```
class Vehicle { // Vehicle: default needed for array declarations,
                // copy for storing in Favorites or Seller
private:
    string type;
    string brand;
    string model;
    int ID;
    int price;
    int year;
    int mileage;
    string condition;
    string fuelType;

    static int totalVehicles;
```

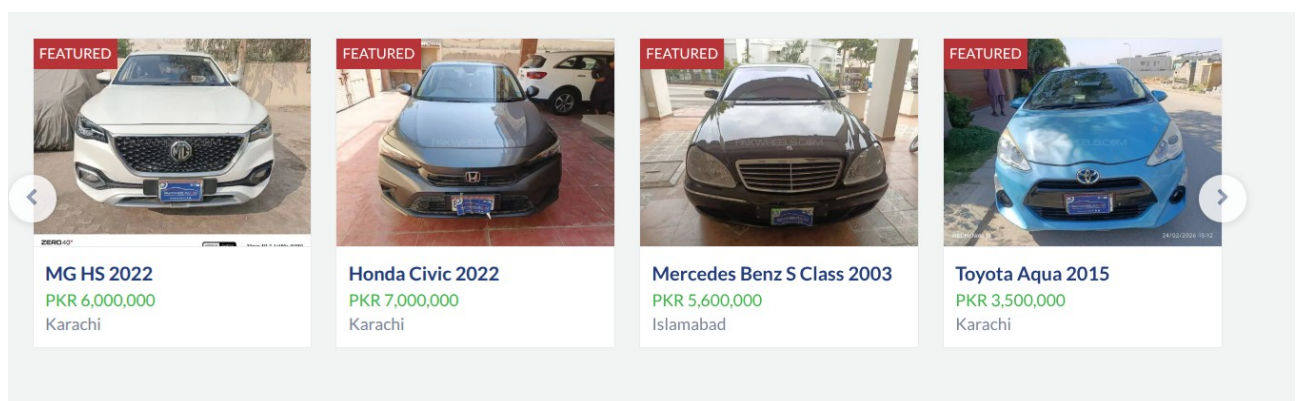
On PakWheels, each car listing shows details like model, brand, price, year, mileage, fuel type, and condition.

Reasoning:

The Vehicle class stores all essential attributes of a car. This ensures encapsulation by grouping related properties and behaviors into a single unit.

Functions like `display()` and `isAffordable()` demonstrate abstraction, allowing interaction without exposing internal data.

Listing Class:



Featured Used Cars for Sale

[View All Featured Used Cars](#)



```
class Listing{ //default needed for listings[] array, parameterized for
private:
    string listingID;
    Vehicle* vehiclePtr; //aggregation
    Seller* const sellerPtr; //aggregation
    string status; //cd be "Pending","Approved","Sold"
    string postDate;
    int price;
public:
```

PakWheels displays approved car advertisements.

Reasoning:

The Listing class links a Vehicle with a Seller and manages status (Pending/Approved).

Pointers are used to show aggregation relationships.
This allows listings to exist independently while referencing existing objects.

Seller Class:

```
class Seller{ // default constructor for array/pointer use, parameterized
private:
    string name;
    string contactInfo;
    string address;
    Vehicle cars[10]; //seller can have up to 10 cars(composition)
    int carCount;
    string ratings;
public:
    Seller():carCount(0) {}
    Seller(string n,string ci,string a,string r):name(n),contactInfo(ci),a
```

Sellers on PakWheels can post multiple
vehicles under their account.

Reasoning:

The Seller class contains an array of Vehicle objects.
This demonstrates **composition**, because vehicles
belong to a seller.

If the seller account is removed, their vehicles
logically disappear from the system.

The class also contains attributes about the seller
such as name,address,contactInfo similar to that in
the website .

Seller Details

● TRUSTED SELLER ●


PAKWHEELS.COM
PAKISTAN'S #1 AUTOMOBILE WEBSITE

Dealer: PakWheels Karachi ✓

Address: Suite No. 303 Third Floor
Tariq Centre Main Tariq
Road

Timings: 09:00 AM to 09:00 PM

[more ads by PakWheels Karachi »](#)



See if your friends know this seller
[Connect with facebook](#)

Message Class:

```
class Message{ // default needed for messages[20] array
private:
    Buyer* senderPtr; //aggregation
    Seller* receiverPtr; //aggregation
    string content;
    string timestamp;
    bool readStatus;
public:
    Message(){}
    Message(Buyer* s,Seller* r,string c,string t,bool
    bool isRead() const { return readStatus; }
    void markAsRead() { readStatus = true; }
    void display() const { cout <<"Message: " << content; }
    void updateContent(string c){content = c; }
};
```

 042XXXXXXXXX
Show Phone Number

 Send Message

Buyers can contact sellers through the platform.

Reasoning:

The Message class connects Buyer and Seller using pointers.

This demonstrates **aggregation**, because messages reference users but do not own them.

It models communication while maintaining separation of concerns.

Review class:

```
}  
class Reviews{ //default for array use, parameteriz  
private:  
    string reviewerName;  
    string reviewText;  
    int ratingScore; // 1 to 5  
    int relatedVehicleID;  
    string relatedSellerName;  
public:  
    Reviews(){}  
    Reviews(string rn,string rt,int rs,int vid,string
```

1 - 12 of 5432 Results Sort By: - Sort By -



close the eyes and buy it
2026 Peugeot 2008 Active
★★★★★
Posted by Ghulam Rasool Paracha on Mar 02, 2026
Familiarity: I owned this car.

“ if we talk about this cars extirior so it's a little bit of triky to use because I am a japanis car owner and every thing is very hard to do like the undigators are up side down if talk about style so I live it's diamond cut design the fuel economy is also very very good and same to wards the performance I don't like the value for money because it's 70 to 78 lacs meanwhile u was having a budget of 60 to70 lacs.close your eyes and buy it ”

Style	★★★★	Comfort	★★★★	Fuel Economy	★★★★
Performance	★★★★	Value for Money	★★★★		

Users can leave reviews or ratings both in text and score as shown in class attributes.

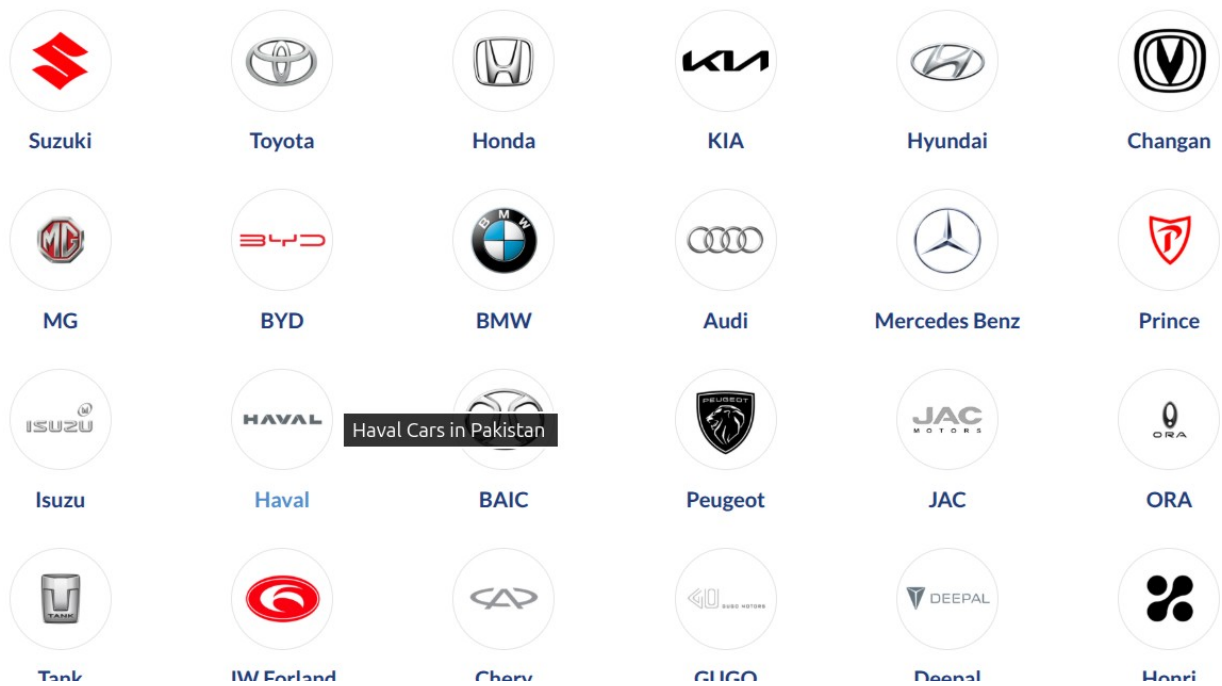
Reasoning:

The Reviews class stores feedback and rating score.

It demonstrates encapsulation by grouping review data together.

This improves transparency in the simulated marketplace.

Brand Class:



```
class Brand{ //default for array use, parameterized
private:
    string brandName;
    string country;
    const int yearFounded;
    string popularModels[5];
    int numberOfCarsListed;
public:
    Brand():yearFounded(0), numberOfCarsListed(0){}
    Brand(string b,string c,int y,int num):brandName
    string getBrandName() const{ return brandName; }
    string getCountry() const{ return country; }
```

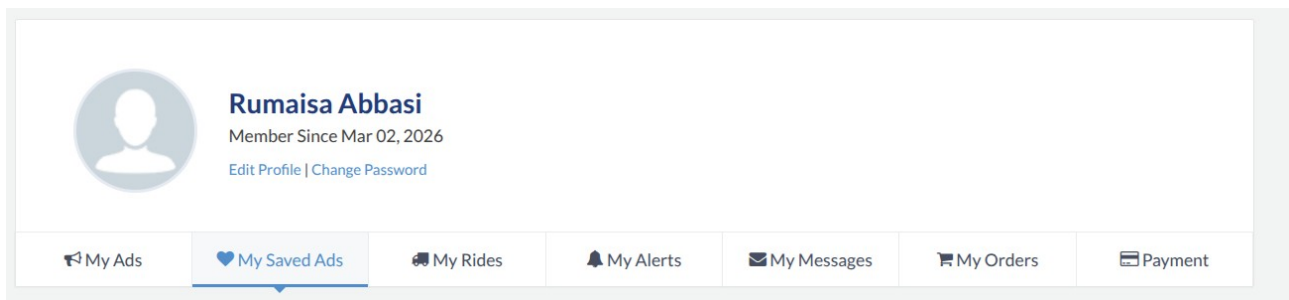
PakWheels categorizes vehicles by brand (Toyota, Honda, Suzuki, etc.). and has a new cars by make section where data abt the brand is provided.

Reasoning:

The brand class allows logical grouping of vehicles by manufacturer. This improves modularity and

makes future expansion easier such as adding brand-based filtering.
This reflects real-world modeling using object-oriented design.

Buyer Class:



```
class Buyer{ //default for array use, parameterized
private:
    string name;
    string contactInfo;
    string address;
    Favorites myFavorites;    //composition
    Message messages[20];    //messages from sellers (
    int messageCount;
public:
    Buyer():messageCount(0) {}
```

Buyers can browse listings, save favorites, and contact sellers similar to the website.

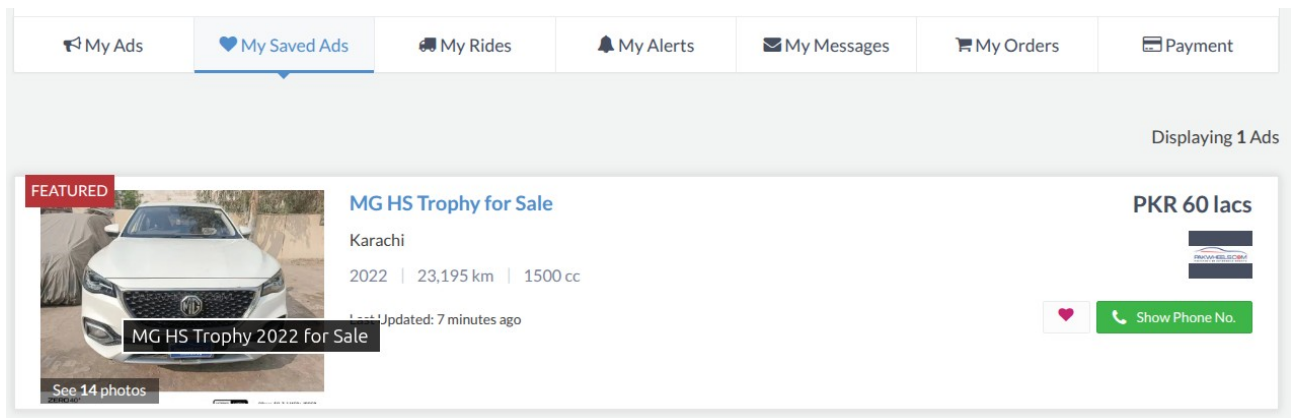
Reasoning:

The Buyer class represents users who interact with listings.

It contains Favourites and Messages objects, demonstrating composition.

This models how a buyer account manages saved cars and communication.

Favourites Class:



```
class Favorites{ //default ctor needed since Buyer is
private:
    string buyerName;
    Vehicle favoriteCars[10];
    int favoriteCount;
    const int maxFavorites;
    string dateOfLastUpdate;
public:
    Favorites():maxFavorites(10),favoriteCount(0){}
    Favorites(string bName,string date):buyerName(bName),dateOfLastUpdate(date){}
    int getFavoriteCount()const {return favoriteCount;}
    void addFavorite(Vehicle* v) {
```

PakWheels allows users to save ads to their favorites list by clicking on the heart button on each listing .

Reasoning:

The Favouritess class stores selected vehicles (My saved ads) .

It is composed inside the Buyer class, meaning favorites cannot exist without a buyer.

This design follows real-world logic and shows strong object ownership (composition) .

Payments Class:

[View Our Products](#)

Available Credits

Cars Featured Ads	0	Bikes Featured Ads	0
Cars Boost Ads	0	Cars Auction Sheet Verification	0
Video Ad Credits	0	Normal Car Ad Credits	0
Normal Bike Ad Credits	0	Normal Accessory Ad Credits	0
Free Normal Car Ad Credits	0	Free Normal Bike Ad Credits	0
Free Normal Accessory Ad Credits	0		

You have not purchased anything from us yet

```
class Payments{ //default for array us
private:
    const string paymentID;
    Listing* listingPtr;
    string buyerName;
    string sellerName;
    string vehicleID;
    int amount;
public:
    Payments(){}
}
```

PakWheels provides online payment services for featured ads, car inspections, and other premium services.

Reasoning:

The Payments class models the transaction process between buyer and seller within the system. It stores details such as payment ID, listing reference, buyer name, seller name, and amount.

A pointer to Listing is used to demonstrate **aggregation**, as the payment is associated with a specific listing but does not own it.

This design reflects how real-world platforms like PakWheels connect financial transactions to specific advertisements while keeping the payment system modular and independent.

OOP Concepts Used :

- Encapsulation - All classes protect their data using private members.
- Abstraction - Functions like display(), approveListing() hide internal logic.
- Composition - Seller–Vehicle, Buyer–Favorites.
- Aggregation - Listing–Vehicle, Message–Users.
- Static Members - Used to track shared system data.
- const Members - Used to ensure certain attributes cannot be modified.